

A PROJECT REPORT

NyayaBot – An AI-Powered Indian Legal Assistant

Submitted In Partial Fulfilment of the
Requirement for the Award of Post
Graduate certificate in Artificial
Intelligence
(PGCP-AI)



Under the Guidance of

(Project Guide)

Submitted By

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CERTIFICATE

CDAC, NOIDA

This is to certify that Report entitled “Natural language processing & AI Based **NyayaBot – An AI-Powered Indian Legal Assistant**” which is submitted by ----- in partial fulfilment of the requirement for the award of Post Graduate certificate in Artificial Intelligence (PGDAI) to CDAC, Noida is a record of the candidates own work carried out by them under my supervision.

The documentation embodies results of original work, and studies are carried out by the student themselves and the contents of the report do not form the basis for the award of any other degree to the candidate or to anybody else from this or any other institution.

(Project Guide)

ABSTRACT

In the modern digital era, access to legal knowledge and professional legal guidance remains a major challenge for a large section of society. Many individuals, especially in rural and semi-urban regions, face difficulties in understanding legal rights, legal procedures, and judicial systems due to lack of awareness, financial limitations, and limited access to legal experts. As a result, people often remain unaware of their legal responsibilities and protections, leading to legal disputes, exploitation, and injustice. With the rapid advancement of Artificial Intelligence, Machine Learning, and Natural Language Processing technologies, intelligent systems can now be developed to provide automated assistance in complex domains such as law. This project, titled “NyayaBot – An AI Based Legal Assistant System”, aims to design and develop an intelligent chatbot that provides instant, reliable, and user-friendly legal guidance through natural language interaction. The system analyses user queries using NLP techniques such as tokenization, semantic analysis, and intent recognition, and retrieves relevant legal information from a structured knowledge base containing statutory laws, case records, and legal guidelines. The chatbot supports both English and Hinglish languages to ensure better accessibility for Indian users and is implemented using modern backend frameworks and cloud-based infrastructure for improved scalability and performance.

The proposed system is designed to assist users in understanding their legal position and available remedies by generating accurate and context-aware responses. Nyaya Bot utilizes pre-trained language models and information retrieval mechanisms to deliver meaningful legal explanations while maintaining data security and privacy. User interaction data is securely stored and analysed to improve system efficiency and response accuracy through continuous learning. The primary objective of the system is not to replace professional legal practitioners but to serve as a preliminary advisory tool that enhances legal awareness and digital empowerment among citizens. Performance evaluation shows that Nyaya Bot provides fast response time, high accuracy in intent classification, and satisfactory user experience under various test conditions. The system effectively handles multiple concurrent user queries and adapts to diverse legal scenarios. In conclusion, Nyaya Bot demonstrates the practical application of Artificial Intelligence in the legal domain and contributes to improving access to justice.

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CHAPTER- 1

1.INTRODUCTION:

The legal system forms the backbone of any democratic society by ensuring justice, equality, and protection of fundamental rights. It establishes rules and regulations that govern individual and institutional behaviour and provides mechanisms for resolving disputes in a lawful manner. In India, the legal framework consists of constitutional provisions, statutory laws, judicial precedents, and administrative regulations. Although these laws are designed to protect citizens, their complexity and technical nature often make them difficult to understand for the general public.

In recent years, rapid growth in population, urbanization, and industrialization has resulted in an increasing number of legal disputes and regulatory requirements. As a consequence, courts and legal institutions are burdened with heavy workloads, leading to delays in justice delivery. For common citizens, accessing legal support becomes time-consuming and financially demanding. Many individuals hesitate to seek legal help due to fear of legal procedures, social stigma, and uncertainty regarding outcomes.

Advancements in Information Technology and Artificial Intelligence have created new opportunities for automating knowledge-based services. Artificial Intelligence enables machines to analyse large volumes of data, recognize patterns, and generate intelligent responses. Natural Language Processing allows computers to understand and interpret human language. These technologies have been successfully applied in various domains such as healthcare, finance, education, and customer service.

NyayaBot is developed as an AI-based legal assistant that utilizes these technologies to provide instant legal guidance. The system allows users to communicate in natural language and obtain relevant legal information in a simplified manner. By combining intelligent algorithms with structured legal databases, NyayaBot aims to enhance accessibility, efficiency, and transparency in legal awareness.

BACKGROUND:

Traditionally, legal services have been provided through lawyers, courts, legal advisors, and government institutions. This system ensures professional expertise and legal accuracy, but it also presents several limitations. High consultation fees, limited availability of legal professionals, and geographical constraints restrict access to legal services for many individuals. Rural and remote areas often lack adequate legal infrastructure, which further increases inequality in access to justice.

In addition, legal language is highly technical and formal, making it difficult for non-lawyers to interpret statutes, judgments, and contracts. Many people depend on informal sources such as acquaintances, social media, or unverified websites for legal advice. Such sources often provide incomplete or misleading information, resulting in wrong decisions and legal complications.

With the introduction of digital governance and e-court systems, several legal services have been moved online. Online case filing, virtual hearings, and digital documentation have improved administrative efficiency. However, these systems still require users to possess basic legal and technical knowledge. As a result, a significant portion of the population remains excluded from digital legal services.

Artificial Intelligence has emerged as a powerful tool for handling complex data and knowledge-intensive tasks. In the legal sector, AI has been used for document classification, legal research, contract analysis, and case outcome prediction. Several international legal platforms have adopted AI for improving productivity and reducing operational costs. NyayaBot builds upon these technological foundations and adapts them to the Indian legal context, focusing on accessibility and simplicity.

OBJECTIVE:

The primary objective of NyayaBot is to develop an intelligent and reliable legal assistance system that provides preliminary legal guidance through conversational interaction. The project aims to simplify complex legal concepts and present them in a form that is easily understandable by common users. The system is designed to understand user queries expressed in natural language and accurately identify the legal intent behind them. By utilizing advanced Natural Language Processing techniques such as intent detection, keyword extraction, and semantic analysis, NyayaBot ensures effective interpretation of user input. In addition, the project focuses on developing a structured and comprehensive legal knowledge base that contains relevant statutes, provisions, case references, and guidelines to support accurate information retrieval.

Another important objective of the project is to integrate Artificial Intelligence models for generating context-aware and meaningful responses that address user queries effectively. The system is developed to support bilingual communication in English and Hinglish in order to enhance accessibility for a wider range of users. Special emphasis is placed on ensuring secure storage and processing of user data, maintaining privacy and confidentiality. The backend architecture is designed to be scalable and suitable for cloud deployment, allowing the system to handle multiple users efficiently. Furthermore, the performance of NyayaBot is evaluated using parameters such as response accuracy, processing speed, and user satisfaction. Through continuous improvement and analysis, the project aims to promote legal awareness, digital literacy, and equitable access to justice among citizens.

MOTIVATION:

The motivation for developing NyayaBot arises from the widespread lack of legal awareness among citizens and the increasing difficulty faced by individuals in accessing affordable and reliable legal guidance. A large section of society remains unaware of their fundamental rights, legal remedies, and procedural requirements, which often leads to exploitation, prolonged disputes, and injustice. Many people hesitate to approach legal professionals due to high consultation fees, fear of complex legal procedures, and uncertainty regarding legal outcomes. As a result, individuals frequently rely on informal sources such as friends, social media platforms, and unverified websites for legal advice, which may provide inaccurate or misleading information. These challenges highlight the urgent need for a reliable and easily

accessible legal assistance system that can bridge the gap between legal institutions and common citizens.

The rapid advancement of Artificial Intelligence and Natural Language Processing technologies has created new opportunities for developing intelligent systems capable of handling complex information efficiently. The success of AI-based chatbots in domains such as healthcare, banking, and customer service served as a strong inspiration for applying similar technologies in the legal sector. Personal interest in emerging technologies, combined with the desire to contribute to social welfare, further motivated the development of this project. NyayaBot represents an effort to utilize technical expertise to address real-world problems and support digital empowerment. By providing instant and reliable legal guidance, the system aims to promote justice, transparency, and legal literacy among users.

PURPOSE AND SCOPE:

The primary purpose of NyayaBot is to function as a preliminary legal advisory platform that assists users in gaining a basic understanding of legal matters before approaching professional legal services. The system is designed to serve as a first point of contact for individuals seeking information about their rights, responsibilities, and legal procedures. By presenting legal concepts in simple and understandable language, NyayaBot helps users overcome confusion and hesitation related to legal processes. It also enables users to identify possible remedies and actions relevant to their specific situations, thereby supporting informed decision-making.

In addition, the project aims to reduce dependency on unreliable and informal sources of legal information by providing accurate and structured guidance. NyayaBot contributes to improving transparency and accountability in legal awareness initiatives and supports government efforts related to digital governance and citizen empowerment. The system also serves as an educational tool for students, researchers, and individuals interested in understanding the application of Artificial Intelligence in the legal domain. Through continuous improvement and system enhancement, the project seeks to establish a sustainable and trustworthy platform for legal assistance.

The scope of the NyayaBot project defines the functional boundaries and operational limitations of the system. The project primarily focuses on providing informational legal assistance based on Indian laws and judicial practices. It covers commonly encountered legal

domains such as civil, criminal, family, consumer, labour, and property laws. The system supports text-based interaction through a web-based platform and is designed to process multiple user queries efficiently. It incorporates cloud-based deployment techniques to ensure scalability, reliability, and availability across different geographical locations.

However, NyayaBot does not provide legally binding advice and does not replace professional legal practitioners or judicial authorities. The system is intended solely for educational and informational purposes and should not be used as a substitute for expert legal consultation. Certain limitations exist with respect to language support, offline accessibility, and advanced case analysis. Future extensions of the project may include regional language support, voice-enabled interaction, mobile application development, integration with court databases, and advanced legal analytics. Thus, the current scope establishes a strong foundation for further research and development in intelligent legal assistance systems.

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2. SURVEY OF TECHNOLOGY

The continuous advancement of Information and Communication Technology has revolutionized traditional systems by transforming them into intelligent, automated, and data-driven platforms. Modern computing systems are now capable of processing vast amounts of information in real time and generating meaningful insights. Technologies such as Artificial Intelligence, Machine Learning, Natural Language Processing, and Cloud Computing have played a crucial role in this transformation. These technologies enable the development of smart applications that can learn from data, adapt to changing environments, and provide personalized services.

The legal sector is characterized by large volumes of unstructured textual data, including laws, judgments, contracts, case files, and government regulations. Manual analysis and interpretation of such information is time-consuming and requires specialized expertise. As the number of legal documents increases continuously, there is a growing need for automated systems that can assist in information processing and retrieval. NyayaBot is developed using a combination of advanced computational techniques and software tools to address these challenges. The system integrates Artificial Intelligence, web technologies, and cloud platforms to provide efficient legal assistance. This chapter presents a comprehensive survey of the technologies employed in the development of NyayaBot and explains how they contribute to system effectiveness, scalability, and reliability.

ARTIFICIAL INTELLIGENCE:

Artificial Intelligence is a multidisciplinary field of computer science that aims to create systems capable of simulating human cognitive functions such as reasoning, learning, perception, and problem-solving. AI systems use mathematical models, algorithms, and computational frameworks to analyse data and generate intelligent outputs. Depending on their functionality, AI systems can be classified into rule-based systems, machine learning-based systems, and deep learning-based systems.

In the context of NyayaBot, Artificial Intelligence serves as the foundation for intelligent decision-making and response generation. The system utilizes AI algorithms to analyse user queries, identify relevant legal contexts, and select appropriate information from the knowledge

base. AI techniques are applied to manage ambiguity in language, handle incomplete queries, and interpret user intentions accurately. By simulating human reasoning processes, NyayaBot can provide explanations that are logical, consistent, and easy to understand. The integration of AI enhances system adaptability and enables continuous improvement through learning mechanisms.

MACHINE LEARNING:

Machine Learning is a subset of Artificial Intelligence that focuses on developing algorithms capable of learning from data and improving their performance over time. Machine Learning models extract patterns and relationships from historical data and use them to make predictions or classifications. Based on learning methodology, Machine Learning techniques can be categorized into supervised learning, unsupervised learning, semi-supervised learning, and reinforcement learning.

NyayaBot employs Machine Learning techniques for multiple tasks, including intent detection, query classification, relevance ranking, and performance optimization. Supervised learning algorithms are trained using labelled datasets containing examples of legal queries and corresponding categories. Unsupervised learning methods are used to identify hidden patterns and clusters in user interaction data. Reinforcement learning techniques can be applied to optimize response selection based on user feedback. By continuously analysing usage patterns, NyayaBot improves its accuracy and adaptability, making the system more reliable over time.

NATURAL LANGUAGE PROCESSING:

Natural Language Processing is a specialized field of Artificial Intelligence that focuses on enabling computers to understand, interpret, and generate human language in a meaningful manner. NLP integrates computational linguistics, statistical analysis, and deep learning to process textual and spoken data.

In NyayaBot, NLP techniques are applied at multiple stages of query processing. Initially, preprocessing operations such as tokenization, normalization, punctuation removal, and stop-word elimination are performed. Stemming and lemmatization techniques are then used to reduce words to their root forms. Higher-level NLP methods such as named entity recognition, part-of-speech tagging, syntactic parsing, and sentiment analysis help in extracting semantic information. The system also incorporates Hinglish processing mechanisms by mapping

commonly used mixed-language expressions into standardized formats. These techniques enable NyayaBot to interpret complex legal queries accurately and generate relevant responses.

LARGE LANGUAGE MODELS:

Large Language Models are advanced deep learning models trained on massive datasets containing diverse textual information. These models use transformer architectures and self-attention mechanisms to capture long-range dependencies and contextual relationships in language. Popular LLMs include GPT, BERT, and T5 families.

NyayaBot integrates Large Language Models to enhance language understanding and response generation capabilities. These models assist in summarizing lengthy legal texts, interpreting statutory provisions, and generating simplified explanations. Pre-trained models are utilized to reduce training time and computational cost. Fine-tuning techniques may be applied using domain-specific legal datasets to improve model performance. LLMs also enable contextual continuity in conversations, allowing NyayaBot to maintain coherence across multiple interactions.

INFORMATION RETRIEVAL SYSTEMS:

Information Retrieval systems are designed to locate and retrieve relevant information from large document collections based on user queries. IR techniques focus on document indexing, query processing, similarity measurement, and ranking.

NyayaBot employs hybrid information retrieval techniques combining traditional keyword-based search with modern vector-based similarity search. Keyword matching methods such as TF-IDF are used for exact term matching, while embedding-based techniques capture semantic similarity. Document embeddings are generated using neural models and stored in vector databases. Similarity measures such as cosine similarity are applied to rank documents. This hybrid approach ensures high precision and recalls in retrieving legal information.

FastAPI FRAMEWORK:

Fast API is a modern, high-performance web framework for building RESTful APIs using Python. It is designed for speed, simplicity, and scalability. Fast API supports asynchronous request handling, automatic data validation, and interactive documentation.

In NyayaBot, Fast API serves as the backbone of the backend system. It handles HTTP requests, manages routing, validates input data, and coordinates communication between

different modules. Asynchronous programming enables efficient resource utilization and improves response time. Fast Api's dependency injection system simplifies code organization and enhances maintainability. Automatic API documentation facilitates testing and debugging during development.

DATABASE AND DATAMANAGMENT:

Data management plays a critical role in storing, organizing, and retrieving information efficiently. Effective data management ensures consistency, reliability, and security of system resources.

NyayaBot utilizes structured data formats such as JSON and lightweight database systems to store legal documents, user queries, and system logs. Indexing mechanisms improve query performance. Data normalization techniques reduce redundancy and improve consistency. Backup and recovery strategies protect against data loss. Encryption and access control mechanisms ensure confidentiality. Proper data governance policies support long-term system sustainability.

CLOUD COMPUTING AND DEPLOYMENT:

Cloud computing provides scalable, flexible, and cost-effective computing resources through virtualization and distributed infrastructure. It enables organizations to deploy applications without investing in physical hardware.

NyayaBot is deployed using containerization technologies such as Docker and hosted on cloud platforms like Hugging Face Spaces. Containers ensure consistent execution environments across development and production systems. Cloud deployment supports automatic scaling, load balancing, and fault tolerance. Continuous integration and deployment pipelines enable automated testing and updates. These features improve system availability and operational efficiency.

FRONTEND TECHNOLOGIES:

Frontend technologies determine the quality of user interaction and overall usability of the system. A well-designed interface enhances accessibility, efficiency, and user satisfaction.

Nyaya Bot's frontend is developed using standard web technologies such as HTML, CSS, and JavaScript. Responsive design ensures compatibility with mobile devices, tablets, and desktops. Interactive elements improve engagement and navigation. Asynchronous API calls

enable real-time data exchange. User experience design principles are applied to minimize cognitive load and simplify interaction.

SECURITY TECHNOLOGIES:

Security is a fundamental requirement for modern web applications, particularly those handling sensitive information. Effective security mechanisms protect systems from unauthorized access, data leakage, and cyber attacks.

NyayaBot incorporates multiple layers of security, including authentication, authorization, encryption, and intrusion detection. API keys are securely managed using environment variables. HTTPS protocols ensure secure communication. Input validation prevents injection attacks. Regular security audits and software updates help identify and mitigate vulnerabilities. These measures ensure system trustworthiness and compliance with data protection standards.

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3.FEASIBILITY STUDY:

A feasibility study is conducted to determine whether the proposed system is practical, efficient, and suitable for successful implementation. For Nyaya Bot, an AI-based legal assistance system, the feasibility study focuses on evaluating technical, economic, and behavioural aspects of the project. This study helps in understanding the availability of required resources, cost-effectiveness, and acceptance of the system by users. By analysing these factors, it becomes possible to assess whether Nyaya Bot can be developed, deployed, and maintained effectively in a real-world environment.

TECHNICAL FEASIBILITY:

The technical feasibility of NyayaBot refers to the availability and suitability of the technologies required for its development and implementation. The system is developed using modern programming languages and frameworks such as Python, Flask or Fast API, and various Natural Language Processing libraries including spaCy, NLTK, and transformer-based models like BERT. These technologies are widely used in the field of artificial intelligence and are supported by extensive documentation and active developer communities. This makes it easier to design, implement, and maintain the system efficiently. The database management system used in Nyaya Bot ensures secure and reliable storage of user queries, legal documents, and case records. The frontend technologies such as HTML, CSS, and JavaScript enable smooth and interactive communication between the user and the system. Moreover, the hardware requirements of the project are minimal, as it can operate on standard personal computers and cloud servers. Advanced hardware such as GPUs may be used for faster model training, but they are not mandatory for basic operations. Therefore, the availability of required software tools, hardware resources, and technical expertise confirms that Nyaya Bot is technically feasible.

ECONOMIC FEASIBILITY:

Economic feasibility examines whether the project can be developed and operated within reasonable financial limits. NyayaBot is economically feasible because it mainly relies on

open-source software tools and freely available machine learning libraries, which significantly reduce development and licensing costs. Programming environments and development platforms used in the project are also available at no cost. Publicly available datasets and legal resources can be utilized for training and testing purposes, minimizing data acquisition expenses. The primary costs involved in the project are related to internet connectivity, basic system hardware, and optional cloud hosting services. Since cloud platforms offer flexible pricing models, the system can be deployed according to budget and usage requirements. Additionally, the modular design of NyayaBot allows future upgrades and expansions without major financial investment. As a result, the project can be implemented at a low cost while still delivering efficient performance, making it economically viable for academic and practical applications.

BEHAVIOURAL FEASIBILITY:

Behavioural feasibility refers to the extent to which the proposed system is accepted and used by its intended users. NyayaBot is designed with a user-friendly interface that allows users to interact with the system through simple and natural language queries. This makes the system easy to use for individuals with different levels of technical knowledge, including students, legal professionals, and general users. The chatbot-based interaction ensures that users receive instant responses without requiring complex procedures. The system reduces dependency on manual legal consultations for basic queries, saving time and effort for users. Proper guidance and help features can be provided within the interface to assist first-time users. Moreover, since the system aims to provide reliable and consistent legal information, users are more likely to trust and adopt it. The automation of routine legal information retrieval also helps legal professionals focus on more complex cases. Therefore, NyayaBot is behaviorally feasible as it is expected to be well accepted and effectively used by its target audience.

CHAPTER -4

REQUIREMENT and ANALYSIS:

The Requirement and Analysis phase is one of the most important stages in software development, as it helps in understanding the needs of users and defining the functional and technical aspects of the system. In this phase, the objectives, limitations, and expectations of the proposed system are carefully studied. For Nyaya Bot, an AI-based legal assistance system, requirement analysis focuses on identifying user requirements, system capabilities, and technical constraints. This process ensures that the final system meets user expectations and operates efficiently. By clearly analysing the problem and identifying the necessary software and hardware resources, the development process becomes more structured and systematic.

PROBLEM DEFINATION

In the present scenario, access to legal information is often time-consuming, complex, and expensive for common people. Many individuals lack basic knowledge about legal procedures, rights, and responsibilities, which makes it difficult for them to seek timely guidance. Traditional methods of legal consultation require physical visits to lawyers, lengthy documentation, and high consultation fees. Moreover, legal documents and case laws are usually written in complex language, making them difficult for non-professionals to understand. As a result, people often hesitate to approach legal authorities or delay seeking help, which may lead to serious consequences.

With the rapid growth of digital technologies and artificial intelligence, there is a need for an intelligent system that can provide quick, reliable, and easy-to-understand legal assistance. Nyaya Bot aims to solve this problem by offering an automated platform where users can ask legal questions in natural language and receive relevant information instantly. The system analyses user queries using NLP techniques and provides appropriate responses based on legal databases and trained models. The main problem addressed by Nyaya Bot is the lack of easily accessible, affordable, and user-friendly legal support for the general public. By providing instant guidance and basic legal awareness, the system helps users make informed decisions and reduces dependency on manual consultations for routine queries.

SOFTWARE REQUIREMENTS:

The software requirements of Nyaya Bot include the tools, platforms, and programming environments necessary for its development and execution. The system is primarily developed using Python due to its simplicity, flexibility, and strong support for artificial intelligence and machine learning applications. Frameworks such as Flask or Fast API are used to build RESTful APIs that handle communication between the frontend and backend. Natural Language Processing is implemented using libraries such as spaCy, NLTK, and transformer-based models from HuggingFace to analyse and understand user queries. Machine learning frameworks like TensorFlow and PyTorch are used for model training and optimization.

For data storage and management, database systems such as MySQL, PostgreSQL, or MongoDB are used to store user information, query logs, and legal datasets. The frontend interface is developed using HTML, CSS, and JavaScript to ensure a responsive and interactive user experience. Development environments such as Visual Studio Code or PyCharm Community Edition are used for coding and debugging. Version control systems like Git and platforms such as GitHub are utilized for source code management and collaboration. The operating system can be Windows, Linux, or macOS, as all required tools are compatible with these platforms. Web browsers such as Google Chrome, Microsoft Edge, or Mozilla Firefox are required for accessing the system.

HARDWARE REQUIREMENTS:

The hardware requirements of Nyayabot are moderate and easily available, making the system suitable for deployment in both academic and practical environments. A standard personal computer or laptop with a minimum of 8 GB RAM and an Intel i5 or equivalent processor is sufficient for development and basic execution. A minimum storage capacity of 256 GB is recommended to store datasets, models, and project files. For better performance, especially during model training and large-scale data processing, systems with higher RAM and faster processors are preferred.

Although not mandatory, a dedicated Graphics Processing Unit (GPU) can be used to speed up machine learning model training and inference. For deployment, the system can be hosted on cloud servers or local servers depending on availability and user demand. A stable internet connection is required for accessing online datasets, APIs, and cloud services.

CHAPTER-5

5. PRELIMINARY MODULE DESCRIPTION:

The preliminary model of Nyayabot represents the initial design framework that defines the structure, functionality, and working mechanism of the system. This model explains how different components of the system interact with each other to process user queries and generate meaningful legal responses. It acts as a blueprint for development and helps in identifying the major modules involved in the system. The preliminary model integrates user interaction, Natural Language Processing, machine learning techniques, and database management to provide efficient legal assistance.

User Interface Module:

The user interface module serves as the primary point of interaction between the user and the system. It is designed to be simple, interactive, and easy to navigate so that users with minimal technical knowledge can use the system effectively. Users can enter their legal queries in natural language through text input fields. The interface also displays system responses in a clear and readable format. Additional features such as help options, guidance messages, and feedback forms are included to improve user experience. This module ensures smooth communication between users and the backend system.

Query Processing Module:

The query processing module is responsible for receiving user input from the interface and preparing it for further analysis. Once a query is submitted, it is transferred securely to the backend server. In this stage, the input text is cleaned and standardized by removing unnecessary symbols, correcting spelling errors, and converting text into a uniform format. Tokenization and normalization techniques are applied to break the input into meaningful components. These preprocessing steps improve the accuracy and efficiency of subsequent NLP operations.

Natural Language Processing Module:

The Natural Language Processing module plays a crucial role in understanding the meaning and intent of user queries. This module applies techniques such as part-of-speech tagging,

named entity recognition, dependency parsing, and semantic analysis to interpret the query. Transformer-based language models such as BERT are used to extract contextual information and identify relationships between words. Based on this analysis, the system determines the legal domain and intent of the user. This enables accurate classification of queries into categories such as criminal law, civil law, family law, or property law.

Knowledge Base and Database Module:

The knowledge base and database module stores structured and unstructured legal information required for answering user queries. This includes case laws, statutes, government regulations, legal articles, and frequently asked questions. User interaction records and query histories are also maintained for system improvement. Database management systems ensure secure storage, fast retrieval, and efficient data handling. Regular updates are performed to keep the legal information current and reliable. This module forms the foundation of the system's knowledge resources.

Information Retrieval and Reasoning Module:

The information retrieval and reasoning module is responsible for searching relevant legal content based on the processed query. Using similarity matching, semantic search, and ranking algorithms, the system identifies the most appropriate documents from the knowledge base. Machine learning models analyze and prioritize the retrieved results according to relevance and confidence level. Logical reasoning techniques are applied to ensure that the provided response aligns with legal principles. This module enhances the accuracy and usefulness of system outputs.

Response Generation Module:

The response generation module converts retrieved legal information into understandable and user-friendly answers. Complex legal terms are simplified to ensure clarity for common users. The system structures responses in a logical manner and may include references to relevant laws and cases. Disclaimers are also added to inform users that the information is for guidance purposes and does not replace professional legal consultation. This module ensures that users receive clear, accurate, and responsible legal assistance.

Learning and Feedback Module:

The learning and feedback module enables continuous improvement of the system. User feedback, interaction data, and response accuracy records are collected and analysed. These records are used to retrain machine learning models and update system rules. The module helps in identifying system weaknesses and improving performance over time. Continuous learning ensures that Nyayabot adapts to changing legal requirements and evolving user needs.

CHAPTER -6

System designing is the process of defining the architecture, components, interfaces, and data flow of a system to satisfy specified requirements. In Nyayabot, system design focuses on organizing various modules such as user interface, query processing, Natural Language Processing, database management, and response generation in a structured manner. A well-planned system design ensures smooth interaction between components, improves system efficiency, and makes maintenance easier. The design of Nyayabot follows a modular approach, where each component performs a specific function and communicates with other modules through well-defined interfaces.

The system is designed in such a way that user queries are processed in multiple stages, starting from input collection to final response generation. This layered approach improves accuracy, reliability, and scalability. The flow of information between different modules is represented using a flowchart, which provides a clear visual understanding of system operations.

System Flowchart Description:

The flowchart of Nyayabot represents the step-by-step working procedure of the system. It shows how user input is processed and transformed into meaningful legal responses. The flowchart helps developers and users understand the logical sequence of operations performed by the system.

Explanation of System Flowchart:

Start

The process begins when the system is activated and ready to accept user input. At this stage, all system modules are initialized, and database connections are established. The chatbot interface becomes available for user interaction.

User Enters Query

In this step, the user enters a legal query in natural language through the web interface. The user may ask questions related to laws, rights, procedures, or legal cases. The system accepts the input in text format and forwards it to the backend server for processing.

Preprocessing

After receiving the query, the system performs preprocessing operations to clean and standardize the input. This includes removing unnecessary symbols, correcting spelling errors, converting text to lowercase, and eliminating stop words. Tokenization and normalization techniques are also applied. These steps help in improving the quality of input for further analysis.

NLP Analysis

In this stage, Natural Language Processing techniques are applied to understand the structure and meaning of the query. The system performs part-of-speech tagging, named entity recognition, and semantic analysis. Transformer-based language models are used to extract contextual information. This helps the system interpret the user's intent accurately.

Intent and Domain Detection

Based on NLP analysis, the system identifies the intent and legal domain of the query. It determines whether the question belongs to criminal law, civil law, family law, property law, labour law, or other legal areas. This classification helps in directing the query to the appropriate knowledge source.

Search Legal Database

After identifying the domain, the system searches the legal database and knowledge base for relevant information. It retrieves case laws, statutes, regulations, and related documents. Advanced search and similarity matching techniques are used to find the most appropriate results.

Relevance Ranking

The retrieved information is then evaluated and ranked according to relevance, accuracy, and reliability. Machine learning algorithms analyse the content and prioritize the most useful results. This step ensures that users receive high-quality and meaningful responses.

Response Generation

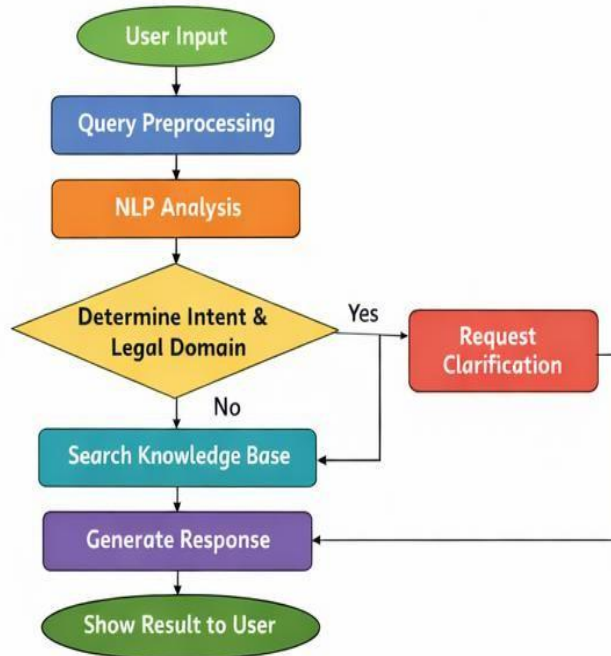
In this step, the system generates a user-friendly response based on ranked information. Complex legal terms are simplified, and the output is structured in a clear and understandable format. References and disclaimers may also be included to maintain legal responsibility.

Display Answer

The generated response is sent back to the user interface and displayed on the screen. The user can read and understand the information easily. This completes the main interaction cycle between the user and the system.

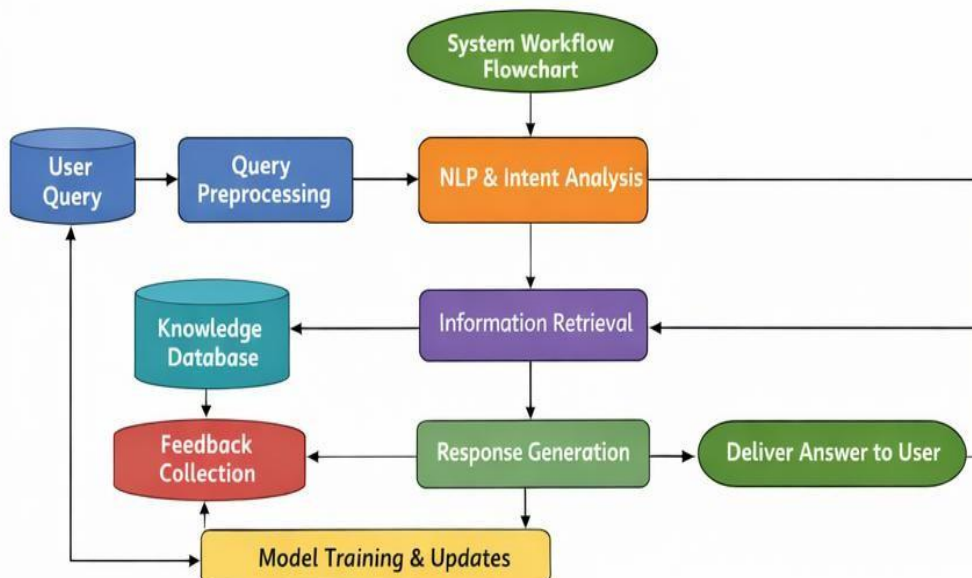
User Query Processing Flowchart

The flowchart below illustrates the process of handling user queries within the system.



System Workflow Flowchart

The flowchart below shows the overall workflow of Nyayabot.



CHAPTER-7

7.CODING:

#openai_client.py

```
import os
import requests
import json

OPENAI_API_KEY = os.getenv("OPENAI_API_KEY")

def generate_answer(prompt: str) -> str:
    try:
        url = "https://api.openai.com/v1/chat/completions"

        headers = {
            "Authorization": f"Bearer {OPENAI_API_KEY}",
            "Content-Type": "application/json"
        }

        payload = {
            "model": "gpt-4o-mini",
            "messages": [
                {
                    "role": "system",
                    "content": (
                        "You are an Indian legal information assistant. "
                        "Explain laws in simple and neutral language. "
                        "Do not give legal advice. "
                        "Do not invent sections."
                    )
                },
                {
                    "role": "user",
                    "content": prompt
                }
            ],
            "temperature": 0.2,
            "max_tokens": 500
        }
    }
```

```

response = requests.post(
    url,
    headers=headers,
    data=json.dumps(payload),
    timeout=30
)

result = response.json()

if "choices" in result:
    return result["choices"][0]["message"]["content"]

if "error" in result:
    return f"OpenAI error: {result['error']['message']}"

return "No response received."

except Exception as e:
    print("OpenAI REST error:", repr(e))
    return "Explanation generation failed."

```

#intent_classifier.py

```

▼ def classify_intent(query: str):

    q = query.lower()

    if any(word in q for word in ["punishment", "jail", "fine", "sentence"]):
        return "punishment_information"

    if any(word in q for word in ["can", "allowed", "legal", "illegal", "valid"]):
        return "legality_check"

    if any(word in q for word in ["how", "procedure", "process", "steps"]):
        return "legal_procedure"

    if any(word in q for word in ["right", "rights", "privacy", "freedom"]):
        return "fundamental_rights"

    if any(word in q for word in ["police", "arrest", "search", "warrant"]):
        return "police_powers"

    return "general_legal_query"

```

#keyword_expander.py

```
✓ KEYWORD_MAP = {
    "police": ["investigating officer", "law enforcement"],
    "phone": ["mobile", "digital device", "electronic device"],
    "arrest": ["custody", "detention"],
    "warrant": ["court order", "judicial permission"],
    "cyber": ["online", "digital crime"],
    "privacy": ["personal liberty", "data protection"]
}

✓ def expand_keywords(query: str) -> str:
    expanded = query
    for word, synonyms in KEYWORD_MAP.items():
        if word in query:
            expanded += " " + " ".join(synonyms)
    return expanded
```

#query_normalizer.py

```
import re

def normalize_query(text: str):
    text = text.lower()
    text = re.sub(r"^[a-z0-9\s]", " ", text)
    return re.sub(r"\s+", " ", text).strip()
```

#translator.py

```
# -----
# Load Hinglish dictionary safely
# -----

BASE_DIR = os.path.dirname(__file__)
HINGLISH_PATH = os.path.join(BASE_DIR, "hinglish_words.json")

with open(HINGLISH_PATH, encoding="utf-8") as f:
    HINGLISH_DICT = json.load(f)

# -----
# Hinglish → English translator
# -----

✓ def translate_hinglish(text: str) -> str:
    """
    Converts common Hinglish words into English
    Example:
    police bina warrant phone check kar sakti hai kya
    → police without warrant phone check can do what
    """

    text = text.lower()

    # normalize punctuation
    text = re.sub(r"[?!.]", "", text)

    words = text.split()
    translated = []

    for w in words:
        translated.append(HINGLISH_DICT.get(w, w))

    return " ".join(translated)
```

#retriever.py

```
DATA_PATH = os.path.join(
    os.path.dirname(__file__),
    "..",
    "data"
)

FILES = [
    "ipc.json",
    "crpc.json",
    "evidence_act.json",
    "it_act.json",
    "constitution.json"
]

def load_documents():
    docs = []
    for f in FILES:
        path = os.path.join(DATA_PATH, f)
        with open(path, encoding="utf-8") as file:
            docs.extend(json.load(file))
    return docs

DOCUMENTS = load_documents()

def retrieve(query: str, top_k=8):
    q = query.lower()
    results = []

    for doc in DOCUMENTS:
        text = (doc.get("title", "") + " " + doc.get("content", "")).lower()
        if any(word in text for word in q.split()):
            results.append(doc)

    return results[:top_k]
```


8.RESULT

**NyayaBot**
Indian Legal Assistant

if someone hit my car what should i do?

Agar koi tumhari car ko hit karta hai, toh pehle kuch zaroori steps lena chahiye:

- **Suraksha pehle**:** Sabse pehle, dekh lo ki tum aur dusre log theek hain ya nahi. Agar koi hurt hua hai, toh turant ambulance bula lo.
- **Car ko safe jagah pe le jao**:** Agar car drive kame layak hai, toh ise roadside ya koi safe jagah pe park kar do.
- **Details collect karo**:** Dusre driver ki details lo, jaise naam, phone number, aur vehicle ka number. Agar possible ho, toh uski insurance details bhi le lo.
- **Photos lo**:** Accident ki photos lo jisme car ka damage aur accident ka scene dikhaye. Yeh sab future mein madad karega.
- **Police ko inform karo**:** Accident ke baare mein police ko batana mat bhoolo. Woh report banaenge jo future mein bahut kaam aayegi.
- **Insurance company se contato karo**:** Apni insurance company ko turant inform karo, taki woh bhi case ko handle kar sakein.
- **Kaaqazaat sambhal ke rakho**:** Saare documents, jaise police report aur insurance papers, sambhal ke rakho.

Zyada tension mat lo, sab kuch theek ho jayega.

Ask in English or Hinglish...

Ask

**NyayaBot**
Indian Legal Assistant

Zyada tension mat lo, sab kuch theek ho jayega.

if someone hit by my car what should i do?

Bot: Agar kisi ko tumhari car se chot lagti hai, toh yeh steps follow karo:

- **Suraksha pehle**:** Sabse pehle check karo ki wo vyakti theek hai ya nahi. Agar koi seriously hurt hua hai, toh turant ambulance bulao.
- **Police ko inform karo**:** Police ko turant call karo, woh situation ko handle karenge aur ek report banayenge.
- **Details le lo**:** Us vyakti ki details lo, jaise naam, contact number, aur jiska accident hua woh track karo. Aur agar koi gawah hain, toh unse bhi information lo.
- **Photos lo**:** Accident ki photos khinch lo, jisse damage dikh sake. Yeh tumhare liye zaroori hoga.
- **Insurance company ko contact karo**:** Apne insurance company ko batao ki accident ho gaya hai, wo aage ki process samjhayenge.
- **Kaaqazaat sambhal ke rakho**:** Sabhi documents, jaise police report aur insurance details, acche se sambhal lo.

Yeh sab karo aur zyada tension mat lo, sab kuch manage ho jayega.

Ask in English or Hinglish...

Ask

what measures should i take?

Bot: Agar tumhe car accident me involve hona padta hai, toh tumhe kuch mahatvapurn measures lena chahiye:

1. ****Suraksha pehle****: Pehle check karo ki tum aur dusre log theek hain ya nahi. Agar koi hurt hai, toh pehle unki madad karo.
 2. ****Emergency services ko call karo****: Police aur ambulance ko bulana mat bhoolo, yeh zaroori hai.
 3. ****Accident ka report****: Police ko detail se batana, jab woh aayein toh unki madad karna. Unka report future me madad karega.
 4. ****Details ka collection****: Dusre driver ki details le lo, jaise naam, phone number, vehicle number, aur insurance details.
 5. ****Gawahon ki jankari****: Agar koi gawah hain, toh unse bhi naam aur contact number le lo. Unki gawahi may future me madad karegi.
 6. ****Photos lena****: Accident ki photos lena na bhoolna, damage, aur accident ka scene dikhate hue.
 7. ****Insurance ko inform karo****: Apne insurance provider ko turant batao ki accident hua hai.
 8. ****Documentation sambhal ke rakhna****: Saare important documents, jaise police report aur insurance papers, achhe se sambhal ke rakhna.
- Yeh steps follow karne se tumhara kaam asaan ho jayega. Stress na lo, sab kuch manage ho jayega!

Ask in English or Hinglish...

Ask

CHAPTER-9

CONCLUSION:

Nyayabot is an AI-based legal assistance system developed to provide quick, reliable, and easy-to-understand legal guidance to users. The system uses Natural Language Processing and machine learning techniques to analyse user queries and generate appropriate responses. It helps users access basic legal information without the need for physical consultation, saving time and cost. The modular design, user-friendly interface, and secure data handling make the system efficient and reliable. Overall, Nyayabot proves to be a practical and effective solution for improving legal awareness and digital legal support.

FUTURE SCOPE:

In the future, Nyayabot can be enhanced by integrating advanced AI models to improve response accuracy and understanding of complex legal queries. Multilingual and voice-based support can be added to increase accessibility. The system can be connected with official legal databases to provide real-time updates. A mobile application can also be developed for easier access. Additional features such as document analysis, personalized guidance, and direct lawyer consultation may further improve system functionality.

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